

M TILAK VENKATA SAI

192424415

DSA0504

15. Keep rows with at least 2 NaN values

```
import pandas as pd
import numpy as np

# Create DataFrame
df = pd.DataFrame({
    'A': [10, np.nan, 30, np.nan, 50],
    'B': [20, np.nan, np.nan, 40, 60],
    'C': [30, 40, np.nan, np.nan, 70],
    'D': [40, 50, 60, 70, np.nan]
})

print("Original DataFrame:")
print(df)

# Keep rows having at least 2 NaN values
result = df[df.isnull().sum(axis=1) >= 2]

print("\nRows with at least 2 NaN values:")
print(result)
```

```
lab 15.py - C:/Users/Tilak/AppData/Local/Programs/Python/Python314/lab 15.py (3.14.2)
File Edit Format Run Options Window Help
import pandas as pd
import numpy as np

# Create DataFrame
df = pd.DataFrame({
    'A': [10, np.nan, 30, np.nan, 50],
    'B': [20, np.nan, np.nan, 40, 60],
    'C': [30, 40, np.nan, np.nan, 70],
    'D': [40, 50, 60, 70, np.nan]
})

print("Original DataFrame:")
print(df)

# Keep rows having at least 2 NaN values
result = df[df.isnull().sum(axis=1) >= 2]

print("\nRows with at least 2 NaN values:")
print(result)

IDLE Shell 3.14.2
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MS(
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
=== RESTART: C:/Users/Tilak/AppData/Local/Programs/Python/Pythor
Original DataFrame:
   A    B    C    D
0 10.0 20.0 30.0 40.0
1  NaN  NaN 40.0 50.0
2 30.0  NaN  NaN 60.0
3  NaN 40.0  NaN 70.0
4 50.0 60.0 70.0  NaN

Rows with at least 2 NaN values:
   A    B    C    D
1  NaN  NaN 40.0 50.0
2 30.0  NaN  NaN 60.0
3  NaN 40.0  NaN 70.0
>>>
```

16. Split DataFrame into groups based on School Code

```
import pandas as pd
```

```
# Create DataFrame
```

```
data = {
```

```
    'school_code': ['S001', 'S002', 'S001', 'S003',
```

```
                  'S002', 'S001', 'S003', 'S002'],
```

```
    'name': ['Arun', 'Priya', 'Rahul', 'Anjali',
```

```
            'Kiran', 'Ravi', 'Neha', 'Amit'],
```

```
    'age': [18, 19, 17, 18, 20, 19, 21, 18]
```

```
}
```

```
df = pd.DataFrame(data)
```

```
print("Original DataFrame:")
```

```
print(df)
```

```
# Group by school code
```

```
grouped = df.groupby('school_code')
```

```
# Display groups
```

```
for code, group in grouped:
```

```
    print("\nSchool Code:", code)
```

```
    print(group)
```

```
# Check type
```

```
print("\nType of GroupBy object:")
```

```
print(type(grouped))
```

Output:

```
import pandas as pd

# Create DataFrame
data = {
    'school_code': ['S001', 'S002', 'S001', 'S003',
                   'S002', 'S001', 'S003', 'S002'],
    'name': ['Arun', 'Priya', 'Rahul', 'Anjali',
            'Kiran', 'Ravi', 'Neha', 'Amit'],
    'age': [18, 19, 17, 18, 20, 19, 21, 18]
}

df = pd.DataFrame(data)

print("Original DataFrame:")
print(df)

# Group by school code
grouped = df.groupby('school_code')

# Display groups
for code, group in grouped:
    print("\nSchool Code:", code)
    print(group)

# Check type
print("\nType of GroupBy object:")
print(type(grouped))
```

```
>>>
Rows with at least 2 NaN values:
   A    B    C    D
1 NaN NaN 40.0 50.0
2 30.0 NaN  NaN 60.0
3 NaN 40.0  NaN 70.0

=== RESTART: C:/Users/Tilak/AppData/Local/Programs/Python/Python314/lab 1
Original DataFrame:
  school_code  name  age
0        S001  Arun   18
1        S002  Priya  19
2        S001  Rahul  17
3        S003  Anjali  18
4        S002  Kiran  20
5        S001   Ravi  19
6        S003  Neha   21
7        S002   Amit  18

School Code: S001
  school_code  name  age
0        S001  Arun   18
2        S001  Rahul  17
5        S001   Ravi  19

School Code: S002
  school_code  name  age
1        S002  Priya  19
4        S002  Kiran  20
7        S002   Amit  18

School Code: S003
  school_code  name  age
3        S003  Anjali  18
6        S003   Neha   21

Type of GroupBy object:
<class 'pandas.core.groupby.generic.DataFrameGroupBy'>
>>>
```

17. School Code-wise Mean, Minimum and Maximum Age

You can use the **same school dataset**.

```
import pandas as pd
```

```
data = {

    'school_code': ['S001', 'S002', 'S001', 'S003',

                   'S002', 'S001', 'S003', 'S002'],
```

```

        'name': ['Arun', 'Priya', 'Rahul', 'Anjali',
                  'Kiran', 'Ravi', 'Neha', 'Amit'],

        'age': [18, 19, 17, 18, 20, 19, 21, 18]
    }

df = pd.DataFrame(data)

print("Original DataFrame:")

print(df)

# Group by school code and calculate age statistics
result = df.groupby('school_code')['age'].agg(

    ['mean', 'min', 'max']

)

print("\nMean, Minimum and Maximum Age:")

print(result)

```

output:

```

import pandas as pd

data = {
    'school_code': ['S001', 'S002', 'S001', 'S003',
                   'S002', 'S001', 'S003', 'S002'],

    'name': ['Arun', 'Priya', 'Rahul', 'Anjali',
             'Kiran', 'Ravi', 'Neha', 'Amit'],

    'age': [18, 19, 17, 18, 20, 19, 21, 18]
}

df = pd.DataFrame(data)

print("Original DataFrame:")
print(df)

# Group by school code and calculate age statistics
result = df.groupby('school_code')['age'].agg(
    ['mean', 'min', 'max']
)

print("\nMean, Minimum and Maximum Age:")
print(result)

```

```

School Code: S001
  school_code  name  age
0      S001  Arun   18
2      S001  Rahul  17
5      S001   Ravi  19

School Code: S002
  school_code  name  age
1      S002  Priya  19
4      S002  Kiran  20
7      S002   Amit  18

School Code: S003
  school_code  name  age
3      S003  Anjali  18
6      S003   Neha  21

Type of GroupBy object:
<class 'pandas.core.groupby.generic.DataFrameGroupBy'>
>>>
=== RESTART: C:/Users/Tilak/AppData/Local/Programs/Python/Python314/lab 1'
Original DataFrame:
  school_code  name  age
0      S001  Arun   18
1      S002  Priya  19
2      S001  Rahul  17
3      S003  Anjali  18
4      S002  Kiran  20
5      S001   Ravi  19
6      S003   Neha  21
7      S002   Amit  18

Mean, Minimum and Maximum Age:
              mean  min  max
school_code
S001           18.0   17   19
S002           19.0   18   20
S003           19.5   18   21

```